



+34 629860362



Barcelona



antonivalls@proton.me



[in/antoni-valls-cifre](#)



[antonivalls.github.io](#)

Computer Skills

Programming: Python, R, Matlab, C++, SQL, Fortran

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, CUDA

Computer Vision: OpenCV, Visual-Inertial Odometry, Depth Estimation, SfM, 3D Reconstruction, Gaussian Splatting.

NLP & Vision-Language: Transformers, OpenAI API, Multi-Modal LLMs

Graph & Networks: NetworkX, Graph Neural Networks

Infrastructure & Tools: Linux, Git, LaTeX, ROS, Slurm, AWS, PowerBI, Blender

Languages

English
Advanced

Italian
Beginner

Catalan and Spanish
Native

References

Jordi Sanchez-Riera
Scientific Researcher (IRI, CSIC-UPC)
[jsanchez@iri.upc.edu](#)

Jesus Cerquides
Scientific Researcher & Learning Systems
Department Head (IIIA-CSIC)
[j.cerquides@csic.es](#)

Antoni Valls

AI RESEARCH | COMPUTER VISION | DEEP LEARNING

About Me

AI researcher and engineer with a background in **theoretical physics and data science**, working on computer vision, visual localization, and vision-language models. Experienced across the full pipeline from model design to real-world deployment.

Work Experience

Robotic Engineer | IRI, CSIC - UPC

Barcelona. April 2025 - Present

- Participation in the SMARTGAZE II project with Biel Glasses, enhancing mobility for people with low vision through **AI-powered smart glasses**.
- Focused on **visual urban localization** in dynamic environments and adaptation of state-of-the-art models for precise navigation with and without GPS.
- First-authored a paper submitted to IEEE T-ITS benchmarking **VLMs models for assisitive navigation** in urban environments. ArXiv [here](#).
- Tools used: **Visual-Inertial Odometry, Multi-Modal Large Language Models, Feature Extraction & Matching, Segmentation Models, Depth Estimation, SfM and 3D Scene Reconstruction, Kalman Filters, GNSS Correction, and Graph Theory.**

Research Assistant | IIIA - CSIC

Barcelona. April 2024 - October 2024

- Co-authored an **NLP research project** on geopolitically-informed BERT-based models for propaganda detection, with results **published as an article**.
- Collaborated on a **computational sociology project** for my master's thesis, investigating the influence of link-prediction algorithms on social contagion models. Responsibilities include dataset collection, exploratory data analysis, coding **GNN models** (e.g., GCN, GAT, SEAL), developing diffusion algorithms, and compiling related research. Code, report, and presentation [here](#).
- Tools used:** **Natural Language Processing, Deep Learning, LLMs, Generative AI, Reinforcement Learning, Graph Neural Networks, Computational Sociology, and Data Analytics.**

IT Risk Advisory Consultant | Deloitte Advisory, S.L.

April 2022 - September 2022

Education and Training

MSc in Data Science | University of Padua

Padua. October 2022 - December 2024

- Curriculum in Machine Learning for Intelligent Systems.
- Skills: **Computer Vision**, CNNs, **OpenAI API**, Transformers, **NLP, LLMs**, and Variational Autoencoders (VAEs). Also Statistics, **Data Analytics and Visualization**.
- Master's thesis: "Exploring the Effects of [Graph](#) Neural Network-Based Link Prediction on Social Contagion Dynamics".

BSc in Theoretical Physics | Universitat de Barcelona

Barcelona. September 2017 - January 2022

- Honors in Linear Algebra and Geometry.
- End-of-degree project on **Nuclear Physics**: "[Atomic and nuclear \$\Xi\$ -bound states](#)".